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Chapter 1

Introduction to differential equations

1.1 Introduction

- In the sciences and engineering, mathematical models are developed to aid in the understanding of physical phenomena.
- These models often yield an equation that contains some derivatives of an unknown function.
- Such an equation is called a '*differential equation*'
- The simplest example of model developed in calculus is the free fall of a body.
  - In this case, an object is dropped from a certain height above the ground and falls under the force of gravity.
  - Newton's second law, which states that object's mass times its acceleration equal the total force acting on it, can be applied to the falling object. This leads to the equation

$$m\frac{d^2h}{dt^2} = -mg$$

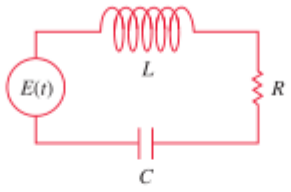
  - This is a differential equation containing the second derivative of the unknown height, h as a function of time.

– Fortunately, the above equation is easy to solve for h. (Divide the equation by m)

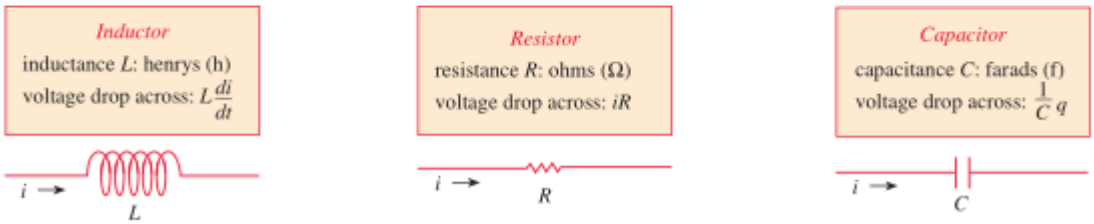
$$\begin{aligned} m\frac{d^2h}{dt^2} &= -mg \\ \frac{d^2h}{dt^2} &= -g \\ \frac{dh}{dt} &= -gt + c_1 \\ h(t) &= -\frac{gt^2}{2} + c_1 t + c_2 \end{aligned}$$

- Diferential equation arise a variety of subject areas, including not only the physical sciences but also such diverse fields as economics, medicine, psychology and operations research.

Example 1. A classical application of differential equations is found in the study of an electric circuit.



(a) *LRC*-series circuit



L is the inductance
R is the resistance
$E(t)$ is the eIectromotive force
$g(t)$ is the charge
t is the time

The application of Kirchoff's laws leads to the equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{c}q = E(t)$$

Example 2. In psychology, one model of the learning of a task involves the equation

$$\frac{dy/dt}{y^{3/2}(1-y)^{3/2}} = \frac{2p}{\sqrt{n}}$$

Here the variable y represents the learner's skill level as a function of time t. The constants p and n depend on the individual learner and the nature of the task.

1.2 Terminology

If an equation involves the derivative of one variable with respect to another, then the former is called a *dependent variable* and the latter an *independent variable*. Thus in the equations :

$$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = 0 \tag{1.1}$$

t is the independent variable
x is the dependent variable
a, b is the coefficients

and

$$\frac{\partial x}{\partial x} - a\frac{\partial u}{\partial y} = x - 2y \tag{1.2}$$

x, y is the independent variable
u is the dependent variable

A differential equation involving ordinary derivatives with respect to a single independent variable is called an *ordinary differential equation*, equation (1.1).

A differential equation involving partial derivatives with respect to more than one independent variable is called an *partial differential equation*, equation (1.2).

The *order* of a differential equation is the order of the highest-order derivatives present in the equation.

The *degree* of a differential equation is the degree of the highest-order derivatives present in the equation.

Example 3.

1. $\frac{d^3u}{dt^3} + 7\frac{du}{dt} + u = \sin t$

2. $5\frac{d^2x}{dy^2} + 7(\frac{d^3x}{dy^3})^5 + 2x = 0$

3. $(\frac{d^4p}{dt^4})^2 + \frac{d^3p}{dt^3} = 6t$

In 1. order is _____ and degree is _____

In 2. order is _____ and degree is _____

In 3. order is _____ and degree is _____

A *linear differential equation* is one in which the occurrence of the dependent variable y has the format

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = F(x)$$

where

1. $a_n(x), a_{n_1}(x), \ldots, a_0(x)$ and $F(x)$ depend only on the independent variable x.

2. the highest degree of the dependent variable y and its derivative is up to 1.

3. there is no multiplication between y and any of its derivatives.

4. there is no transcendental function of y or its derivatives.

Example 4.

1. $\frac{d^2y}{dx^2} + y^3 = 0$

2. $\frac{d^2y}{dx^2} - y^3 = x^3$

3. $\frac{d^2y}{dx^2} + y\frac{dy}{dx} = \cos x$

4. $y' + \frac{1}{x}y = x^3 - 3$

5. $x' + \frac{3}{y}x = 2y$

6. $\frac{dy}{dx} = a(x)y + b(x)$

7. $(1 + y^2)\frac{d^2y}{dt^2} + t\frac{dy}{dt} + y = e^y$

1.3 Solution and Initial Value Problems

An nth order ordinary differential equation is an equality involving the independent variable x, the dependent variable y, and the first n derivatives of y.

Example 5.

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = x^3 \quad (2^{\text{nd}} \text{ order})$$
$$1 - \left(\frac{d^3y}{dx^3}\right)^2 - y = 0 \quad (3^{\text{rd}} \text{ order})$$
$$\frac{d^4y}{dx^4} = xy \quad (4^{\text{th}} \text{ order})$$

Thus a general form for an nth order equation would be

$$F\left(x, y, \frac{dy}{dx}, \ldots, \frac{d^ny}{dx^n}\right) = 0 \tag{1.3}$$

where

- F is a function of the independent variable x, the dependent variable y, and the derivatives of y up to order n; that is, $x, y, \ldots, \frac{d^ny}{dx^n}$.
- We assume that the equation holds for all x in an interval I, $[a, b]$ or $(a, b]$ or (a, b).

In many cases, we can isolate the highest-order term $\frac{d^ny}{dx^n}$, and write as

$$\frac{d^ny}{dx^n} = f\left(x, y, \frac{dy}{dx}, \ldots, \frac{d^{n-1}y}{dx^{n-1}}\right). \tag{1.4}$$

1.4 Explicit solution

A function $\phi(x)$ that when subtituted for y in equation (1.3) and (1.4) and satisfies the equation for all x in the interval I is called an *explicit solution* to the equation on I.

Example 6. Show that $\phi(x) = x^2 - x^{-1}$ is an explicit solution to

$$\frac{d^2y}{dx^2} - \frac{2}{x^2}y = 0 \tag{1.5}$$

Solution. *Differentiating,*

$$\begin{aligned}\phi(x) &= x^2 - x^{-1}\\ \phi'(x) &= 2x + x^{-2}\\ \phi''(x) &= 2 - 2x^{-3}.\end{aligned}$$

Substitute $\phi'(x)$ *and* $\phi''(x)$ *into* (1.5),

$$\begin{aligned}(2 - 2x^{-3}) - \tfrac{2}{x^2}(x^2 - x^{-1}) &= 2 - 2x^{-3} - 2 + 2x^{-3}\\ &= 0.\end{aligned}$$

Therefore, $\phi(x) = x^2 - x^{-1}$ *is an explicit solution to* (1.5) *on* $(-\infty, 0) \cup (0, \infty)$.

Example 7. Show that for any choice of the constants c_1 and c_2, the function $\phi(x) = c_1 e^{-x} + c_2 e^{2x}$ is an explicit solution to

$$y'' - y' - 2y = 0 \tag{1.6}$$

Solution. *Differentiating,*

$$\begin{aligned}\phi(x) &= c_1 e^{-x} + c_2 e^{2x}\\ \phi'(x) &= -c_1 e^{-x} + 2c_2 e^{2x}\\ \phi''(x) &= c_1 e^{-x} + 4c_2 e^{2x}.\end{aligned}$$

Substitute $\phi'(x)$ *and* $\phi''(x)$ *into* (1.6),

$$\begin{aligned}c_1 e^{-x} + 4c_2 e^{2x} - (-c_1 e^{-x} + 2c_2 e^{2x}) - 2(c_1 e^{-x} + c_2 e^{2x})&\\ = (c_1 + c_1 - 2c_1)e^{-x} + (4c_2 - 2c_2 - 2c_2)e^{2x}&\\ = 0.&\end{aligned}$$

Therefore, $\phi(x) = c_1 e^{-x} + c_2 e^{2x}$ *is an explicit solution to* (1.6) *on* $(-\infty, \infty)$ *for any choice of the constant* c_1 *and* c_2.

1.5 Implicit solution

A relation $G(x, y) = 0$ is said to be an implicit solution to equation (1.4) on the interval I if it defines one or more explicit solutions on I.

Example 8. Show that $y^2 - x^3 + 8 = 0$ is an implicit solution to

$$\frac{dy}{dx} = \frac{3x^2}{2y} \tag{1.7}$$

on the interval $(2, \infty)$.

Solution.

$$\begin{aligned} y^2 - x^3 + 8 &= 0 \\ y &= \pm\sqrt{x^3+8}, \\ \frac{dy}{dx} &= \pm\frac{3x^2}{2\sqrt{x^3+8}}. \end{aligned}$$

Both $y(x)$ and $y'(x)$ are defined on $(2,\infty)$. Substitute them into (1.7),

$$\text{LHS}: \quad \frac{3x^2}{2\sqrt{x^3+8}} = \frac{3x^2}{2\sqrt{x^3+8}} \quad : \text{RHS}$$

(You can check that $y(x) = \sqrt{x^3+8}$ is also an implicit solution to (1.7).

Note : We hereafter use the term 'solution' to mean an explicit or implicit solution.

1.6 Initial Value Problem

By an '*initial value problem*' for an nth order differential equation

$$F\left(x, y, \frac{dy}{dx}, \ldots, \frac{d^n y}{dx^n}\right) = 0$$

We mean: Find a solution to the differential equation on an interval I that satisfies at x_0 the n initial conditions.

$$\begin{aligned} y(x_0) &= y_0 \\ \frac{dy}{dx}(x_0) &= y_1 \\ &\vdots \\ \frac{d^{n-1}y}{dx^{n-1}}(x_0) &= y_{n-1} \end{aligned}$$

where $x_0 \in I$ and $y_0, y_1, \ldots, y_{n-1}$ are given constants.

In the case of a first order equation, the initial conditions reduce to the single requirement

$$y(x_0) = y_0,$$

and in the case of a second order equation, the initial conditions have the form

$$y(x_0) = y_0, \quad \frac{dy}{dx}(x_0) = y_1.$$

Example 9. Show that $y(x) = \sin x - \cos x$ is a solution to the initial value problem

$$\frac{d^2y}{dx^2} + y = 0, \tag{1.8}$$

where $y(0) = -1$ and $\frac{dy}{dx}(0) = 1$.

Solution. *Differentiating,*

$$\begin{aligned} y(x) &= \sin x - \cos x \\ y'(x) &= \cos x + \sin x \\ y''(x) &= -\sin x + \cos x \end{aligned}$$

Substitute $y(x)$ and $y''(x)$ into (1.8)

$$\begin{aligned} \frac{d^2y}{dx^2} + y &= (-\sin x + \cos x) + (\sin x - \cos x) \\ &= 0. \end{aligned}$$

Therefore, $y(x)$ is a solution to (1.8) on the interval $(-\infty, \infty)$.

When we check the initial conditions, we find

$$\begin{aligned} y(0) &= \sin 0 - \cos 0 = -1, \\ y'(0) &= -\sin 0 + \cos 0 = 1, \end{aligned}$$

which meets the requirements of (1.8). Therefore $y(x)$ is a solution to the given initial value problem.

Example 10. Given the general solution for the initial value problem

$$y'' - y' - 2y = 0, \quad y(0) = 2, \quad y'(0) = -3$$

is $y(x) = c_1e^{-x} + c_2e^{2x}$, find the value of c_1 and c_2.

Solution. *Differentiating,*

$$\begin{aligned} y(x) &= c_1e^{-x} + c_2e^{2x} \\ y'(x) &= -c_1e^{-x} + 2c_2e^{2x}, \end{aligned}$$

and substituting,

$$\begin{aligned} y(0) = 2: \quad c_1e^{-(0)} + c_2e^{2(0)} &= 2 \\ c_1 + c_2 &= 2, \qquad (1.9) \\ y'(0) = -3: \quad -c_1e^{(0)} + 2c_2e^{2(0)} &= -3 \\ -c_1 + 2c_2 &= -3. \qquad (1.10) \end{aligned}$$

Solving (1.9) and (1.10) we obtain $c_2 = \frac{1}{3}$ and $c_1 = \frac{7}{3}$.

Therefore $y(x) = \frac{7}{3}e^{-x} - \frac{1}{3}e^{2x}$ is a particular solution.

Chapter 2

Separable equations

The general form of a linear first order differential equation is

$$\frac{dy}{dx} = f(x, y)$$

The equation is separable if it is of the form

$$\frac{dy}{dx} = \underbrace{g(x)}_{\text{depends only on } x} \cdot \underbrace{h(y)}_{\text{depends only on } y}$$

A separable equation can be expressed as

$$\frac{1}{h(y)}dy = g(x)dx,$$

and therefore

$$\int \frac{1}{h(y)}dy = \int g(x)dx + c.$$

Example 11. Solve $\dfrac{dx}{dt} = 3xt^2$.

Solution. *The equation is separable. Upon separating the variables and integrating we obtain*

$$\int \frac{1}{3x}dx = \int t^2 dt$$
$$\frac{1}{3}\ln x = \frac{t^3}{3} + c.$$

Example 12. Solve $\dfrac{dy}{dx} = \dfrac{ye^{x+y}}{x^2+2}$.

Solution. *Separating the variables and integrating,*

$$\frac{dy}{dx} = \frac{ye^x e^y}{x^2+2}$$
$$\int \frac{1}{ye^y} dy = \int \frac{e^x}{x^2+2} dx.$$

Example 13. Solve the initial value problem

$$\frac{dy}{dx} = \frac{y-1}{x+3}, \quad y(-1) = 0.$$

Solution. *This is the separable equation, so*

$$\int \frac{1}{y-1}\, dy = \int \frac{1}{x+3}\, dx,$$
$$\ln|y-1| = \ln|x+3| + c. \tag{2.1}$$

Substituting $x = -1, y = 0$ *into* (2.1) *we obtain*

$$\ln|0-1| = \ln|-1+3| + c$$
$$\ln 1 = \ln 2 + c$$
$$0 = \ln 2 + c$$
$$c = -\ln 2.$$

Therefore (2.1) *becomes*

$$\ln|y-1| = \ln|x+3| - \ln 2$$
$$\ln|y-1| = \ln\frac{|x+3|}{2}$$
$$|y-1| = \frac{|x+3|}{2}$$
$$|y-1| = \frac{x+3}{2} \quad \text{or} \quad y-1 = -\frac{x+3}{2}.$$

Therefore $y = -\frac{x+5}{2}$ *or* $y = -\frac{x+1}{2}$. *Now* $y = -\frac{x+5}{2}$ *does not satisfy the initial value condition because* $0 = y(-1) \neq -\frac{-1+5}{2} = 2$ *and* $y = -\frac{x+1}{2}$ *satisfies the initial value condition because* $0 = y(-1) = -\frac{-1+1}{2} = 2$.

Thus the solution (explicit solution) to the initial value problem is $y = -\frac{1}{2}(x+1)$.

Chapter 3

Linear first order equations

A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \tag{3.1}$$

is said to be a **linear equation** in the dependent variable y. When $g(x) = 0$, the linear equation (3.1) is said to be **homogeneous**; otherwise, it is **nonhomogeneous**.

By dividing both sides of (3.1) by the lead coefficient $a_1(x)$ we obtain the **standard form** of a linear equation

$$\frac{dy}{dx} + p(x)y = q(x).$$

3.1 The homogeneous DE

The homogeneous equation

$$\frac{dy}{dx} + p(x)y = 0 \tag{3.2}$$

is separable. By writing (3.2) as

$$\frac{dy}{y} + p(x)\,dx = 0,$$

and integrating, and solving for y gives

$$y = ce^{-\int p(x)\,dx}.$$

3.2 The nonhomogeneous DE

Let

$$\frac{dy}{dx} + p(x)y = q(x) \tag{3.3}$$

be a nonhomogeneous equation.

Observe that

$$\frac{d}{dx}\left(e^{\int p(x)\,dx}\right) = p(x)\, e^{\int p(x)\,dx},$$

and

$$\frac{d}{dx}\left(e^{\int p(x)\,dx} \cdot y\right) = e^{\int p(x)\,dx}\,\frac{dy}{dx} + p(x)\, e^{\int p(x)\,dx}\, y. \tag{3.4}$$

Now if we multiply (3.3) by $e^{\int p(x)\,dx}$, called the **integrating factor**, equation becomes

$$e^{\int p(x)\,dx}\frac{dy}{dx} + p(x)e^{\int p(x)\,dx}y = e^{\int p(x)\,dx}q(x) \tag{3.5}$$

and (3.5) can be expressed as

$$\frac{d}{dx}\left(e^{\int p(x)\,dx} \cdot y\right) = e^{\int p(x)\,dx}q(x)$$

by (3.4). This implies

$$e^{\int p(x)\,dx} \cdot y = \int e^{\int p(x)\,dx}q(x)\,dx + c$$

for some constant c, and it follows

$$y = e^{-\int p(x)\,dx} \cdot \int e^{\int p(x)\,dx}q(x)\,dx + e^{-\int p(x)\,dx} \cdot c.$$

Example 14. Find the general solution to $\dfrac{dy}{dx} + y = 2 + 2x$.

Solution. *Here $p(x) = 1$ and $q(x) = 2+2x$. Compute the integrating factor: $\mu(x) = e^{\int 1\,dx} = e^x$. Multiply the equation by the integrating factor and solve*

$$\begin{aligned}
e^x \cdot \frac{dy}{dx} + e^x \cdot y &= e^x(2 + 2x) \\
\frac{d}{dx}(e^x \cdot y) &= e^x(2 + 2x) \\
e^x \cdot y &= \int e^x(2 + 2x)\,dx \\
&= 2e^x + 2xe^x - 2e^x + c \\
&= 2xe^x + c \\
y &= 2x + ce^{-x}.
\end{aligned}$$

Example 15. Find the general solution to $\dfrac{1}{x}\dfrac{dy}{dx} - \dfrac{2y}{x^2} = x\cos x, \quad x > 0.$

Solution. *Multiply the equation by x to obtain*

$$\frac{dy}{dx} - \frac{2}{x}y = x^2\cos x.$$

Here $p(x) = -\frac{2}{x}$ and $q(x) = x^2\cos x$. Compute the integrating factor: $\mu(x) = e^{\int -2/x\,dx} = e^{-2\ln|x|} = e^{\ln 1/x^2} = \frac{1}{x^2}$. Multiply the equation by the integrating factor and solve

$$\begin{aligned}
\frac{1}{x^2}\frac{dy}{dx} - \frac{2}{x^3}y &= \cos x \\
\frac{d}{dx}\left(\frac{1}{x^2}\cdot y\right) &= \cos x \\
\frac{1}{x^2}\cdot y &= \int \cos x\,dx \\
&= \sin x + c \\
y &= x^2\sin x + cx^2.
\end{aligned}$$

Example 16. Solve $\dfrac{dy}{dx} + xy = 4x.$

Solution. *Integrating factor is $\mu(x) = e^{\int x\,dx} = e^{x^2/2}$. Multiply the equation by the integrating factor and solve*

$$\begin{aligned}
e^{x^2/2}\cdot\frac{dy}{dx} + e^{x^2/2}\cdot y &= e^{x^2/2}(4x) \\
\frac{d}{dx}(e^{x^2/2}\cdot y) &= 4xe^{x^2/2} \\
e^{x^2/2}\cdot y &= \int 4xe^{x^2/2}\,dx \\
&= 4e^{x^2/2} + c \\
y &= 4 + ce^{-x^2/2}.
\end{aligned}$$

Example 17. Solve $\dfrac{du}{dv} + \dfrac{1}{v}u = v^3, \quad v > 0.$

Solution. *Integrating factor is $\mu(v) = e^{\int 1/v\,dx} = e^{\ln|v|} = e^{\ln v} = v$ since*

$v > 0$. *Multiply the equation by the integrating factor and solve*

$$\begin{aligned} v\frac{du}{dv} + u &= v^4 \\ \frac{d}{dv}(u \cdot v) &= v^4 \\ (u \cdot v) &= \int v^4 \, dv \\ &= \frac{v^5}{5} + c \\ u &= \frac{v^4}{5} + \frac{c}{v}. \end{aligned}$$

Example 18. Solve

$$t \cos z \, dz + \sin z \, dt = 0.$$

Solution.

$$\begin{aligned} \sin z \, dt &= -t \cos z \, dz \\ \frac{dt}{dz} &= -t\frac{\cos z}{\sin z} \\ \frac{dt}{dz} + t\frac{\cos z}{\sin z} &= 0. \end{aligned}$$

Integrating factor is $\mu(x) = e^{\int \cot z \, dz} = e^{\ln|\sin z|} = |\sin z|$. *Multiply the equation by the integrating factor and solve*

$$\begin{aligned} |\sin z|\frac{dt}{dz} + |\sin z|\frac{\cos z}{\sin z}t &= 0 \\ \frac{d}{dz}(t \cdot |\sin z|) &= 0 \\ t \cdot |\sin z| &= c \\ t &= \frac{c}{|\sin z|}. \end{aligned}$$

Note that by definition $|\sin z| = [(\sin z)^2]^{1/2}$. *The derivative is* $\frac{1}{2}[(\sin z)^2]^{-1/2} \cdot 2 \sin z \cdot \cos z = \dfrac{\sin z \cos z}{|\sin z|}$ *and* $\dfrac{\sin z}{|\sin z|} = \dfrac{|\sin z|}{\sin z}$.

Chapter 4

Bernoulli's equation

The differential equation

$$\frac{dy}{dx} + p(x)y = q(x)y^n \tag{4.1}$$

where n is any real number, is called **Bernoulli's equation**.[1] Note that for $n = 0$ and $n = 1$, equation (4.1) is linear. For $n \neq 0$ and $n \neq 1$, the substitution $u = y^{1-n}$ reduces any equation of the form (4.1) to a linear equation.

Example 19. Solve $\dfrac{dy}{dx} - \dfrac{1}{2x}y = 5x^2y^5$.

Solution. *Make the substitution* $v = y^{1-5} = y^{-4}$. *Then* $\dfrac{dv}{dx} = -4y^{-5}\dfrac{dy}{dx}$. *Multiply the equation by* $-4y^{-5}$,

$$\begin{aligned} -4y^{-5} \cdot \frac{dy}{dx} - -4y^{-5} \cdot \frac{1}{2x}y &= -4y^{-5} \cdot 5x^2y^5 \\ -4y^{-5} \cdot \frac{dy}{dx} + \frac{2}{x}y^{-4} &= -20 \cdot 5x^2 \\ \frac{dv}{dx} + \frac{2}{x}v &= -20x^2. \end{aligned}$$

Use the method for solving linear equation. The integrating factor is

$$\mu(x) = e^{\int 2/x\, dx} = e^{2\ln x} = e^{\ln x^2} = x^2,$$

[1] Name after the Swiss mathematician Jacob Bernoulli (1654–1705)

multiply the equation by the integrating factor,

$$\begin{aligned} x^2 \cdot \frac{dv}{dx} + x^2 \cdot \frac{2}{x} v &= x^2 \cdot -20x^2 \\ x^2 \frac{dv}{dx} + 2xv &= -20x^4 \\ \frac{d}{dx}(x^2 \cdot v) &= -20x^4 \\ x^2 \cdot v &= \int -20x^4 \\ x^2 \cdot v &= -20\frac{x^5}{5} + c \\ v &= -4x^3 + \frac{c}{x^2} \\ y^{-4} &= -4x^3 + \frac{c}{x^2}. \end{aligned}$$

Example 20. Solve $\dfrac{dy}{dx} = xy^2 + 2xy$.

Solution. *Rewriting, the equation is a Bernoulli's equation*

$$\frac{dy}{dx} - 2xy = xy^2,$$

with $n = 2$. *Make the substitution* $v = y^{1-2} = y^{-1}$. *Then* $\dfrac{dv}{dx} = -y^{-2}\dfrac{dy}{dx}$. *Multiply the equation by* $-y^{-2}$,

$$\begin{aligned} -y^{-2} \cdot \frac{dy}{dx} - -y^{-2} \cdot 2xy &= -y^{-2} \cdot xy^2 \\ -y^{-2} \cdot \frac{dy}{dx} + 2xy^{-1} &= -x \\ \frac{dv}{dx} + 2xv &= -x. \end{aligned}$$

The integrating factor is

$$\mu(x) = e^{\int 2x\, dx} = e^{x^2},$$

multiply the equation by the integrating factor,

$$\begin{aligned} e^{x^2}\frac{dv}{dx} + 2xe^{x^2}v &= -xe^{x^2} \\ \frac{d}{dx}(e^{x^2}\cdot v) &= -xe^{x^2} \\ (e^{x^2}\cdot v) &= \int -xe^{x^2}\,dx \\ &= -\frac{1}{2}e^{x^2} + c \\ v &= -\frac{1}{2}e^{x^2} + ce^{-x^2} \\ y^{-1} &= -\frac{1}{2}e^{x^2} + ce^{-x^2}. \end{aligned}$$

Chapter 5

Exact equations

Suppose the differential equation

$$\frac{dy}{dx} = f(x, y)$$

can be rewritten in the differential form

$$M(x, y)\, dx + N(x, y)\, dy = 0, \tag{5.1}$$

If the left hand side of equation (5.1) can be identified as a total differential

$$\begin{aligned} M(x, y)\, dx + N(x, y)\, dy &= \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \\ &= dF(x, y), \end{aligned}$$

then its solutions are given (implicitly) by

$$F(x, y) = C$$

for an arbitrary constant C. Therefore, the differential form

$$M(x, y)\, dx + N(x, y)\, dy$$

is said to be exact in a rectangle R if there is a function $F(x, y)$ such that

$$\begin{aligned} \frac{\partial F(x, y)}{\partial x} &= M(x, y) \\ \text{and} \quad \frac{\partial F(x, y)}{\partial y} dx &= N(x, y) \end{aligned}$$

for all (x, y) in R. That is, the total differential of $F(x, y)$ satisfies

$$dF(x, y) = M(x, y)\, dx + N(x, y)\, dy.$$

If $M(x,y)\,dx + N(x,y)\,dy$ is an exact differential form, then the equation

$$dF(x,y) = M(x,y)\,dx + N(x,y)\,dy = 0$$

is called an exact equation.

Theorem 5.1 (Test For Exactness)**.** *Suppose the first partial derivatives of $M(x,y)$ and $N(x,y)$ are continuous in a rectangle R. Then*

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is an exact equation in R if and only if the compatibility condition

$$\frac{\partial M}{\partial y}(x,y) = \frac{\partial N}{\partial x}(x,y)$$

holds for all (x,y) in R.

Example 21. Determine the exactness of the differential equation $\dfrac{dy}{dx} = \dfrac{3x^2 - y}{x - 1}$.

Solution. *Write the equation in differential form*

$$\begin{aligned}(x-1)\,dy &= (3x^2 - y)\,dx\\ -(3x^2 - y)\,dx + (x-1)\,dy &= 0\\ (y - 3x^2)\,dx + (x-1)\,dy &= 0\end{aligned}$$

Here $M(x,y) = y - 3x^2$ and $N(x,y) = x - 1$. By using Theorem 5.1,

$$\frac{\partial M}{\partial y}(x,y) = 1 = \frac{\partial N}{\partial x}(x,y),$$

therefore equation is exact.

Another way is to rearrange the equation which is in differential form

$$\begin{aligned}(y - 3x^2)\,dx + (x-1)\,dy &= 0\\ y\,dx - 3x^2\,dx + x\,dy - dy &= 0\\ (y\,dx + x\,dy) - 3x^2\,dx - dy &= 0\\ d(xy) - d(x^3) - dy &= 0\\ d(xy - x^3 - y) &= 0\end{aligned}$$

where $F(x,y) = xy - x^3 - y$. Then

$$\frac{\partial F(x,y)}{\partial x}(x,y) = y - 3x^2 = M(x,y),$$

and

$$\frac{\partial F(x,y)}{\partial y}(x,y) = x - 1 = N(x,y),$$

so equation is exact.

This method is a bit harder than using the theorem.

5.1 Solving an exact equation

Determine whether it is an exact equation or not, if YES, then

$$\text{there exists} \quad \frac{\partial F(x,y)}{\partial x} = M(x,y) \tag{5.2}$$

$$\text{and} \quad \frac{\partial F(x,y)}{\partial y} = N(x,y). \tag{5.3}$$

Then integrate (5.2) with respect to x:

$$F(x,y) = \int M(x,y)\, dx + g(y). \tag{5.4}$$

Take the partial derivative with respect to y on both sides of equation (5.4).

$$\frac{\partial F(x,y)}{\partial y} = \frac{\partial}{\partial y}\int M(x,y)\, dx + \frac{\partial}{\partial y} g(y) \tag{5.5}$$

Then compare (5.3) with (5.5) to obtain $g'(y)$. Integrate to obtain $g(y)$ and substitute $g(y)$ into (5.5) to obtain $F(x,y)$. Solution is

$$F(x,y) = c.$$

Example 22. Solve the differential equation $(y - 3x^2)\, dx + (x - 1)\, dy = 0$.

Solution. *Here*

$$\frac{\partial M}{\partial y}(x,y) = 1, \quad \textit{and} \quad \frac{\partial N}{\partial x}(x,y) = 1,$$

and so equation is exact.

There exists $F(x,y)$ such that

$$\frac{\partial F}{\partial x}(x,y) = M(x,y) = y - 3x^2 \tag{5.6}$$

and

$$\frac{\partial F}{\partial y}(x,y) = N(x,y) = x - 1. \tag{5.7}$$

Choose (5.6) and integrate it with respect to x:

$$\begin{aligned} F(x,y) &= \int y - 3x^2\, dx + g(y) \\ &= yx - x^3 + g(y). \end{aligned} \tag{5.8}$$

Take the partial derivative with respect to y of both sides of equation (5.8).

$$\frac{\partial F}{\partial y}(x,y) = x + g'(y). \tag{5.9}$$

Compare (5.7) with (5.9) to obtain $g'(y) = -1$. It follows that $g(y) = -y$. Substitute $g(y) = -y$ into (5.8) to obtain $F(x,y) = yx - x^3 - y$.

The solution is $F(x,y) = yx - x^3 - y = c$.

Example 23. Solve the differential equation

$$(2xy - \sec^2 x)\, dx + (x^2 + 2y)\, dy = 0.$$

Solution. *Here $M(x,y) = 2xy - \sec^2 x$ and $N(x,y) = x^2 + 2y$ and*

$$\frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial y} = 2x,$$

so the equation is exact.

There exists $F(x,y)$ such that

$$\frac{\partial F}{\partial x}(x,y) = M(x,y) = 2xy - \sec^2 x \tag{5.10}$$

and

$$\frac{\partial F}{\partial y}(x,y) = N(x,y) = x^2 + 2y. \tag{5.11}$$

Choose (5.10), then

$$\begin{aligned} F(x,y) &= \int 2xy - \sec^2 x\, dx + g(y) \\ &= x^2 y - \tan x + g(y), \end{aligned} \tag{5.12}$$

then

$$\frac{\partial F(x,y)}{\partial y} = x^2 + g'(y). \tag{5.13}$$

Compare (5.11) with (5.13) to obtain $g'(y) = 2y$ so that $g(y) = y^2$. Substitute $g(y) = y^2$ into (5.12) to obtain $F(x,y) = x^2 y - \tan x + y^2$.

Thus solution is $F(x,y) = x^2 y - \tan x + y^2 = c$.

5.2 Solving a nonexact equation

Suppose the equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is not exact.

Sometimes it is possible to find an **integrating factor** $\mu(x,y)$ so that after multiplying, the left hand side of

$$\mu(x,y)M(x,y)\,dx + \mu(x,y)N(x,y)\,dy = 0 \tag{5.14}$$

is an exact differential. By Theorem 5.1, equation (5.14) is exact if and only if

$$\begin{aligned} \frac{\partial(\mu M)}{\partial y} &= \frac{\partial(\mu N)}{\partial x} \\ \mu\frac{\partial M}{\partial y} + \frac{\partial \mu}{\partial y}M &= \mu\frac{\partial N}{\partial x} + \frac{\partial \mu}{\partial x}N, \qquad \text{by the product rule.} \end{aligned} \tag{5.15}$$

We make a simplifying assumption by taking μ to be a function of one variable.

Suppose μ is a function of x. Then $\dfrac{\partial \mu}{\partial x} = \dfrac{d\mu}{dx}$ and $\dfrac{\partial \mu}{\partial y} = 0$ and so (5.15) can be written as

$$\frac{d\mu}{dx} = \frac{\partial M/\partial y - \partial N/\partial x}{N}\mu = \frac{M_y - N_x}{N}\mu \tag{5.16}$$

If $\frac{M_y - N_x}{N}$ turns out to depend solely on the variable x, then (5.16) is a separable, linear first order differential equation. It follows that

$$\mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx}.$$

Likewise if μ is a function of y

$$\frac{d\mu}{dy} = \frac{\partial N/\partial x - \partial M/\partial y}{M}\mu = \frac{N_x - M_y}{M}\mu \tag{5.17}$$

and it follows that

$$\mu(y) = e^{\int \frac{N_x - M_y}{M}\,dy}.$$

Example 24. Show that $\mu(x,y) = xy^2$ is an integrating factor for

$$(2y - 6x)\,dx + (3x - 4x^2y^{-1})\,dy = 0$$

Use this integrating factor to solve the equation.

Solution. *Here* $M(x,y) = 2y - 6x$ *and* $N(x,y) = 3x - 4x^2y^{-1}$ *and*

$$\frac{\partial M}{\partial y} = 2 \neq \frac{\partial N}{\partial x} = 3 - 8xy^{-1}$$

so equation is not exact.

Multiply the equation by the integrating factor $\mu(x,y) = xy^2$ *to obtain*

$$\begin{aligned}(xy^2)(2y - 6x)\,dx + (xy^2)(3x - 4x^2y^{-1})\,dy &= 0\\ (2xy^3 - 6x^2y^2)\,dx + (3x^2y^2 - 4x^3y)\,dy &= 0\end{aligned}$$

and

$$\frac{\partial(\mu M)}{\partial y} = 6xy^2 - 12x^2y = \frac{\partial(\mu N)}{\partial x}$$

and so the equation is exact.

Then there exists $F(x,y)$ *such that*

$$\frac{\partial F(x,y)}{\partial x} = \mu M(x,y) = 2xy^3 - 6x^2y^2$$

and

$$\frac{\partial F(x,y)}{\partial y} = \mu N(x,y) = 3x^2y^2 - 4x^3y.$$

Integrating with respect to x*,*

$$\begin{aligned}F(x,y) &= \int 2xy^3 - 6x^2y^2\,dx + g(y)\\ &= x^2y^3 - 2x^3y^2\,dx + g(y).\end{aligned}$$

Differentiating $F(x,y)$ *with respect to* y *and comparing it with* $\mu N(x,y)$ *implies* $g'(y) = 0$ *and so* $g(y) = k$*.*

Then $F(x,y) = x^2y^3 - 2x^3y^2 + k$ *and the solution is* $x^2y^3 - 2x^3y^2 + k = c$ *or*

$$x^2y^3 - 2x^3y^2 = c'.$$

for some constant c'*.*

Example 25. Show that $\mu(x,y) = \dfrac{1}{1+x^2y^2}$ is an integrating factor for the differential equation

$$(1 + x^2y^2 + y)\,dx + x\,dy = 0.$$

Use this integrating factor to solve the equation.

Solution. *Here* $M(x,y) = 1 + x^2y^2 + y$ *and* $N(x,y) = x$ *and*

$$\frac{\partial M}{\partial y} = 2x^2y + 1 \neq \frac{\partial N}{\partial x} = 1$$

so equation is not exact.

Multiply the equation by the integrating factor $\mu(x,y) = \frac{1}{1+x^2y^2}$ *to obtain*

$$\frac{1}{1+x^2y^2} \cdot (1 + x^2y^2 + y)\, dx + \frac{1}{1+x^2y^2} \cdot x\, dy = 0.$$
$$\left(1 + \frac{y}{1+x^2y^2}\right)\, dx + \left(\frac{x}{1+x^2y^2} \cdot x\right)\, dy = 0.$$

and

$$\frac{\partial(\mu M)}{\partial y} = \frac{1 - x^2y^2}{(1+x^2y^2)^2} = \frac{\partial(\mu N)}{\partial x}$$

and so the equation is exact.

Then there exists $F(x,y)$ *such that*

$$\frac{\partial F(x,y)}{\partial x} = \mu M(x,y) = 1 + \frac{y}{1+x^2y^2}$$

and

$$\frac{\partial F(x,y)}{\partial y} = \mu N(x,y) = \frac{x}{1+x^2y^2}.$$

Integrating with respect to x*,*

$$\begin{aligned} F(x,y) &= \int 1 + \frac{y}{1+x^2y^2}\, dx + g(y) \\ &= x + \int \frac{1}{1+(xy)^2}\, d(xy) + g(y) \\ &= x + \tan^{-1}(xy) + g(y). \end{aligned}$$

Differentiating $F(x,y)$ *with respect to* y *to obtain*

$$\frac{\partial F(x,y)}{\partial y} = 0 + \frac{x}{1+(xy)^2} + g'(y)$$

and comparing it with $\mu N(x,y)$ *implies* $g'(y) = 0$ *and so* $g(y) = k$*.*

Then $F(x,y) = x + \tan^{-1}(xy) + k$ *and the solution is* $x + \tan^{-1}(xy) + k = c$ *or*

$$x + \tan^{-1}(xy) = c'$$

for some constant c'*.*

Example 26. Solve $(2x^2+y)\,dx+(x^2y-x)\,dy=0$ by finding an integrating factor as a function of x.

Solution. *Here* $M(x,y)=2x^2+y$ *and* $N(x,y)=x^2y-x$ *and*

$$\frac{\partial M}{\partial y}=1\neq\frac{\partial N}{\partial x}=2xy-1,$$

equation is not exact.

By (5.16)*, compute*

$$\begin{aligned}\frac{M_y-N_x}{N}&=\frac{1-(2xy-1)}{x^2y-x}\\&=\frac{2-2xy}{x^2y-x}\\&=\frac{2(1-xy)}{-x(1-xy)}\\&=-\frac{2}{x},\end{aligned}$$

and the integrating factor is

$$\mu(x)=e^{\int -2/x\,dx}=e^{-2\ln x}=e^{\ln 1/x^2}=\frac{1}{x^2}.$$

Multiply the equation by the integrating factor to obtain

$$\begin{aligned}\frac{1}{x^2}\cdot(2x^2+y)\,dx+\frac{1}{x^2}\cdot(x^2y-x)\,dy&=0\\\left(2+\frac{y}{x^2}\right)\,dx+\left(y-\frac{1}{x}\right)\,dy&=0.\end{aligned}$$

Here

$$\frac{\partial(\mu M)}{\partial y}=\frac{1}{x^2}=\frac{\partial(\mu N)}{\partial x}$$

and so the equation in now exact.

Solve to obtain the solution $F(x,y)=2x-\frac{y}{x}+\frac{y^2}{2}=c$.

Example 27. Solve $(2xy)\,dx-(x^2+y^2-1)\,dy=0$ by finding a suitable integrating factor.

Solution. *Here* $M(x,y)=2xy$ *and* $N(x,y)=x^2+y^2-1$

$$\frac{\partial M}{\partial y}=2x\neq\frac{\partial N}{\partial x}=-2x,$$

equation is not exact.

Compute

$$\begin{aligned}\frac{M_y - N_x}{N} &= \frac{1}{x^2+y^2-1}\cdot\big(2x-(-2x)\big)\\ &= \frac{4x}{x^2+y^2-1}, \quad \textit{not a function of solely } x\end{aligned}$$

and compute

$$\begin{aligned}\frac{N_x - M_y}{M} &= \frac{1}{2xy}\cdot\big(-2x-2x)\big)\\ &= \frac{-4x}{2xy}\\ &= -\frac{2}{y}.\end{aligned}$$

Choose the function of y as the integrating factor so

$$\mu(y) = e^{\int -2/y\,dx} = e^{-2\ln y} = e^{\ln 1/y^2} = \frac{1}{y^2}.$$

Multiply the equation by the integrating factor to obtain

$$\begin{aligned}\frac{1}{y^2}\cdot(2xy)\,dx + \frac{1}{y^2}\cdot(x^2+y^2-1)\,dy &= 0\\ \left(\frac{2x}{y}\right)\,dx + \left(\frac{x^2}{y^2}+1-\frac{1}{y^2}\right)\,dy &= 0.\end{aligned}$$

Here

$$\frac{\partial(\mu M)}{\partial y} = -\frac{2x}{y^2} = \frac{\partial(\mu N)}{\partial x}$$

and so the equation in now exact.

Solve to obtain the solution $F(x,y) = \dfrac{x^2}{y} - y - \dfrac{1}{y} = c.$

Chapter 6

Homogeneous equations

Definition 1. A function $f(x, y)$ is called a homogeneous with degree of n if

$$f(\lambda x, \lambda y) = \lambda^n f(x, y)$$

for all λ.

Example 28. The function $f(x, y) = x^4 - x^3 y$ is homogeneous with degree 4 because

$$\begin{aligned} f(\lambda x, \lambda y) &= (\lambda x)^4 - (\lambda x)^3(\lambda y) \\ &= \lambda^4 x^4 - \lambda^4 x^3 y \\ &= \lambda^4 (x^4 - x^3 y) \\ &= \lambda^4 f(x, y). \end{aligned}$$

Example 29. The function $f(x, y) = \dfrac{y - xe^{x/y}}{x}$ is homogeneous with degree 0 because

$$\begin{aligned} f(\lambda x, \lambda y) &= \frac{(\lambda y) - (\lambda x)e^{(\lambda x)/(\lambda y)}}{(\lambda x)} \\ &= \frac{\lambda(y - xe^{x/y})}{(\lambda x)} \\ &= \frac{y - xe^{x/y}}{x} \\ &= \lambda^0 f(x, y). \end{aligned}$$

Theorem 6.1. *If $M(x, y)$ and $N(x, y)$ homogenous with the same degree, then the function*

$$f(x, y) = \frac{M(x, y)}{N(x, y)}$$

is homogeneous with degree zero.

Proof. If $M(x,y)$ and $N(x,y)$ homogeneous with degree n, then

$$M(\lambda x, \lambda y) = \lambda^n M(x,y) \quad \text{and} \quad N(\lambda x, \lambda y) = \lambda^n N(x,y).$$

Then

$$f(\lambda x, \lambda y) = \frac{M(\lambda x, \lambda y)}{N(\lambda x, \lambda y)} = \frac{\lambda^n M(x,y)}{\lambda^n N(x,y)} = \frac{M(x,y)}{N(x,y)} = \lambda^0 f(x,y).$$

□

Theorem 6.2. *If a function $f(x,y)$ is homogeneous with degree of zero, $f(x,y)$ can be expressed as a function of the ratio $\frac{y}{x}$ alone.*

Proof. Let $v = \frac{y}{x}$, so $vx = y$. Then

$$\begin{aligned} f(x,y) &= f(x, vx), && y = vx \\ &= x^0 f(1,v), && f(x,y) \text{ homogeneous with degree of zero} \\ &= f(1,v) \\ &= \text{a function that depends only on } v. \end{aligned}$$

□

Suppose $M(x,y)dx + N(x,y)dy = 0$ are homogeneous with same degree. Then

$$\begin{aligned} N(x,y)dy &= -M(x,y)dx \\ \frac{dy}{dx} &= -\frac{M(x,y)}{N(x,y)} \\ &= f(x,y) \\ &= F(\tfrac{y}{x}). && \text{by Theorem (6.2)} \end{aligned}$$

If $v = \frac{y}{x}$ then $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Then

$$\begin{aligned} v + x\frac{dv}{dx} &= F(v) \\ \frac{dv}{dx} &= \frac{F(v) - v}{x} \\ \int \frac{1}{F(v) - v} dv &= \int \frac{1}{x} dx. \end{aligned}$$

6.1 Solving a homogeneous equation

The equation

$$M(x,y)dx + N(x,y)dy = 0$$

is called a homogeneous equation if both $M(x,y)$ and $N(x,y)$ are homogeneous with the same degree.

To solve the differential equation, we make the substitution $y = vx$. Then $dy = xdv + vdx$. Then the equation becomes separable, which can be integrated.

Example 30. Solve $(x-y)dx + xdy = 0$.

Solution. *Here $M(x,y) = x - y$ and $N(x,y) = x$. Now*

$$\begin{aligned} M(\lambda x, \lambda y) &= (\lambda x) - (\lambda y) = \lambda^1(x-y), \\ N(\lambda x, \lambda y) &= (\lambda x) = \lambda^1(x). \end{aligned}$$

From the definition, M and N are homogeneous with the same degree, so it is a homogeneous equation.

Make the substitution $y = vx$. Then $dy = vdx + xdv$. Equation becomes

$$\begin{aligned} \big(x - (vx)\big)dx + x(vdx + xdv) &= 0 \\ (x - vx + vx)dx + x^2dv &= 0 \\ xdx + x^2dv &= 0 \\ x^2dv &= -xdx \\ \int dv &= \int -\frac{1}{x}dx \\ v &= -\ln|x| + c. \end{aligned}$$

Exercise: Show that the equation can also be solved by using the exact equation method.

Example 31. Solve $(xy + y^2 + x^2)dx - x^2dy = 0$.

Solution. *Here $M(x,y) = xy - y^2 + x^2$ and $N(x,y) = -x^2$. Now*

$$\begin{aligned} M(\lambda x, \lambda y) &= (\lambda x)(\lambda y) - (\lambda y)^2 + (\lambda x)^2 \\ &= \lambda^2(xy) - \lambda^2 y^2 + \lambda^2 x^2 \\ &= \lambda^2(xy + y^2 + x^2) \\ &= \lambda^2 M(x,y), \qquad \textit{homogeneous of degree 2} \end{aligned}$$

and

$$\begin{aligned} N(\lambda x, \lambda y) &= -(\lambda x)^2 \\ &= -\lambda^2 x^2 \\ &= \lambda^2(-x^2) \\ &= \lambda^2 N(x,y), \qquad \textit{homogeneous of degree 2.} \end{aligned}$$

Make the substitution $y = vx$. Then $dy = vdx + xdv$. Equation becomes

$$\begin{aligned} \big(x(vx) + (vx)^2 + x^2\big)dx - x^2\big(vdx + xdv\big) &= 0 \\ (vx^2 + v^2x^2 + x^2 - vx^2)dx - x^3dv &= 0 \\ (v^2x^2 + x^2)dx - x^3dv &= 0 \\ x^2(v^2+1)dx - x^3dv &= 0 \\ x^2(v^2+1)dx &= x^3dv \\ \int \frac{1}{x}dx &= \frac{1}{v^2+1}dv \\ c + \ln|x| &= \tan^{-1}(v) \\ c + \ln|x| &= \tan^{-1}(\tfrac{y}{x}) \\ \tfrac{y}{x} &= \tan(\ln|x| + c) \\ y &= x\tan(\ln|x| + c). \end{aligned}$$

Chapter 7

Differential equations with linear coefficients

We have used various substitutions for y to transform the original equation into a new one that we can solve.

In some cases, we must transform both x and y into new variables, let say u and v. This is the situation for equations with linear coefficients, that is equation of the form

$$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$$

where $a_1, a_2, b_1, b_2, c_1, c_2$ are constants.

7.1 Solving differential equations with linear coefficients

Example 32. Solve $(x - 2y + 1)dx + (4x - 3y - 6)dy = 0$.

Consider the ratios

$$\frac{a_2}{a_1} = \frac{4}{1} = 4, \quad \frac{b_2}{b_1} = \frac{-3}{-2} = \frac{3}{2}.$$

This is Case 1.

Solve for (h, k) that satisfies

$$\begin{aligned} h - 2k + 1 &= 0 \\ 4h - 3k - 6 &= 0, \end{aligned}$$

to obtain $h = 3, k = 2$. Substitute

$$\begin{aligned} x &= X + 3 \\ y &= Y + 2 \end{aligned}$$

into the equation to obtain

$$\begin{aligned} \big[(X+3) - (Y+2) + 1\big]dX + \big[4(X+3) - 3(Y+2) - 6\big]dY &= 0 \\ (X - 2Y)dX + (4X - 3Y)dY &= 0 \qquad (7.1) \end{aligned}$$

The resulting equation is a homogeneous equation. Substitute $Y = vX$ and $dY = vdX + Xdv$ into (7.1),

$$\begin{aligned} (X - 2vX)\, dX + (4X - 3vX)(vdX + Xdv) &= 0 \\ (X - 2vX + 4vX - 3v^2X)\, dX + (4X^2 - 3vX^2)\, dv &= 0 \\ (X + 2vX - 3v^2X)\, dX + (4 - 3v)X^2\, dv &= 0 \\ (1 + 2v - 3v^2)\, dX + (4 - 3v)X\, dv &= 0 \\ (4 - 3v)X\, dv &= -(1 + 2v - 3v^2)\, dX \\ \int \frac{4 - 3v}{1 + 2v - 3v^2}\, dv &= -\int \frac{1}{X}\, dX. \end{aligned}$$

Upon integrating,

$$\begin{aligned} X^4|3v + 1|^5 &= |v - 1|A, & A &= e^{4c} \\ X^4|3\tfrac{Y}{X} + 1|^5 &= |\tfrac{Y}{X} - 1|A, & v &= \frac{Y}{X} \\ |3Y + X|^5 &= |Y - X|A \\ |3(y-2) + (x-3)|^5 &= |(y-2) - (x-3)|A, \quad \text{where } X = x - 3, Y = y - 2 \\ |3y + x - 9|^5 &= A|y - x + 1|. \end{aligned}$$

Example 33. Solve $(x + 2y + 3)dx + (2x - 3y - 6)dy = 0$.

Consider the ratios

$$\frac{a_2}{a_1} = \frac{2}{1} = 2, \quad \text{and} \quad \frac{b_2}{b_1} = \frac{4}{2} = 2.$$

This is Case 2.

Let $z = a_1x + b_1y$ so that $z = x + 2y$ and $dz = dx + 2dy$ or $dy = \frac{dz - dx}{2}$.

Substitute into the equation to obtain

$$
\begin{aligned}
(z+3)\,dx + (2z-1)\left(\frac{dz-dx}{2}\right) &= 0 \\
(2z+6)\,dx + (2z-1)(dz-dx) &= 0 \\
(2z+6-2z+1)\,dx + (2z-1)\,dz &= 0 \\
7dx + (2z-1)\,dz &= 0 \\
\int (2z-1)\,dz &= \int -7\,dx \\
z^2 - z &= -7x + c \\
(x+2y)^2 - (x+2y) + 7x &= c, \qquad \text{where } z = x+2y \\
x^2 + 4xy + 4y^2 + 6x - 2y &= c.
\end{aligned}
$$

Chapter 8

Applications

8.1 Mixing Problems

A problem for which the one-compartment system provides a useful representation is the mixing of fluids in a tank.

Let $x(t)$ represent the amount of a substance in a tank (compartment) at time, t.

To use the compartmental analysis model, we must be able to determine the rates at which this substance enters and leaves the tank.

In mixing problems, one is often given the rate at which a fluid containing the substance flows into the tank, along with the concentration of the substance in that fluid.

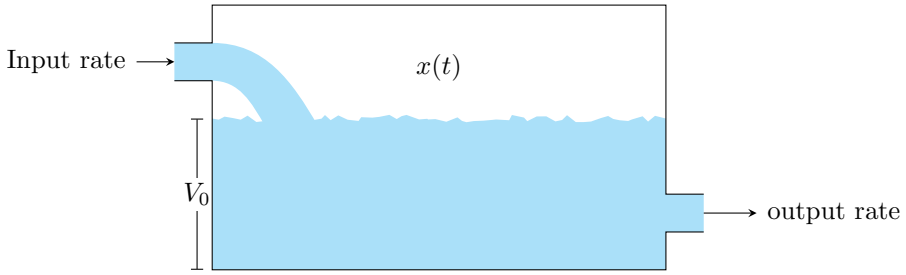
Hence, multiplying the flow rate (volume/time) by the concentration (amount/volume) yields the input rate (amount/time).

The output rate of the substance is usually more difficult to determine.

If we are given the exit rate of the mixture of fluids in the tank, then how do we determine the concentration of the substance in this mixture?

One simplifying assumption that we might make is that the concentration is kept uniform in the mixture. Then we can compute the concentration of the substance in the mixture by dividing the amount $x(t)$ by the volume of the mixture in the tank at time, t.

Multiplying this concentration by the exit rate of the mixture then gives the desired output rate of the substance.



The problem: What is the amount of a substance in a tank at time t ($x(t) = ?$).

$$\begin{aligned}
\frac{dx}{dt} &= \text{changing rate} \\
&= \underbrace{\text{input rate}}_{\text{amount/time: kg min}^{-1}} - \underbrace{\text{output rate}}_{\text{amount/time: kg min}^{-1}} \\
&= \underbrace{(\text{concentration of the substance})}_{\text{amount/volume: kg L}^{-1}} \cdot \underbrace{(\text{enter rate})}_{\text{volume/time: L min}^{-1}} \\
&\quad - \underbrace{(\text{concentration of the substance in the mix})}_{\text{amount in the tank (mix) } x(t)\text{/volume in tank: kg L}^{-1}} \cdot \underbrace{(\text{exit rate})}_{\text{volume/time: L min}^{-1}}
\end{aligned}$$

Example 34. Consider a large tank holding 1000L of water into which a brine solution of salt begins to flow at a constant rate of 6 L min^{-1}. The solution inside the tank is kept well stirred and is flowing out of the tank at a rate of 6 L min^{-1}. If the concentration of salt in the brine entering the tank is 1 kg L^{-1}, determine when the concentration of salt in the tank will reach concentration $\frac{1}{2}$ kg L^{-1}.

Solution. *Let $x(t)$ be the amount of salt at time t. Then*

$$\begin{aligned}
\frac{dx}{dt} &= \textit{changing rate} \\
&= \textit{input rate} - \textit{output rate} \\
&= \textit{(concentration of salt)} \cdot \textit{(enter rate)} \\
&\qquad - \left(\frac{\textit{amount of salt in the tank}}{\textit{volume of fluid in the tank}} \right) \cdot \textit{(exit rate)} \\
&= (1\ kg\ L^{-1}) \cdot (6\ L\ min^{-1}) - \left(\frac{x(t)\ kg}{1000\ L} \right) \cdot 6\ L\ min^{-1} \\
&= 6\ kg\ min^{-1} - \frac{6x}{1000}\ kg\ min^{-1}.
\end{aligned}$$

Therefore, with initial condition $x(0) = 0$,

$$\begin{aligned}\frac{dx}{dt} &= 6 - \frac{6x}{1000} \\ &= \frac{3000 - 3x}{500} \\ &= \frac{3(1000 - x)}{500}.\end{aligned}$$

The resulting equation is separable. Integrating,

$$\begin{aligned}\int \frac{1}{1000 - x}\, dx &= \int \frac{3}{500}\, dt \\ -\ln|1000 - x| &= \frac{3}{500}t + c \\ \ln|1000 - x| &= -\frac{3}{500}t - c \\ 1000 - x &= e^{-3t/500} \cdot e^{-c} \\ x &= 1000 - Ae^{-3t/500}, \quad \textit{where } A = e^{-c}.\end{aligned}$$

By the initial condition $x(0) = 0$, *so* $0 = 1000 - Ae^0$ *which implies* $A = 1000$. *Therefore*

$$\begin{aligned}x(t) &= 1000(1 - e^{-3t/500}) \\ &= \textit{amount salt in the tank at time } t\end{aligned}$$

and

$$\frac{x(t)}{1000} = \textit{the concentration of salt in the tank at time } t.$$

To determine when the concentration of salt is $\frac{1}{2}$ *kg L*$^{-1}$, *we set the right-hand side equal to* $\frac{1}{2}$ *and solve for* t.

$$\begin{aligned}1 - e^{-3t/500} &= \tfrac{1}{2} \\ \tfrac{1}{2} &= e^{-3t/500} \\ \ln\tfrac{1}{2} &= -\frac{3t}{500} \\ \ln 1 - \ln 2 &= -\frac{3t}{500} \\ t &= \frac{500 \ln 2}{3} \approx 115.5 \textit{ min.}\end{aligned}$$

If

$$\textit{enter rate} \neq \textit{exit rate}$$

then

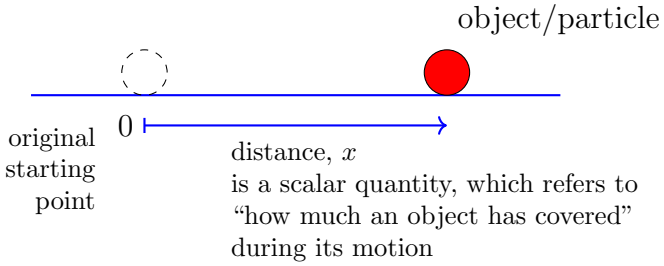
$$\frac{dx}{dt} = (1\ kg\ L^{-1}) \cdot (6\ L\ min^{-1}) - \left(\frac{x(t)\ kg}{(1000 + t)\ L}\right) \cdot (5\ L\ min^{-1}).$$

The difference between the rate of flow into the tank and the rate of flow out is $6 - 5 = 1L\ min^{-1}$. *Therefore, the volume of fluid in the tank after* t *minutes is* $(1000 + t)L$.

8.2 Kinematics

Kinematics is the science of decribing the motion of object without regard to the causes of the motion.

In this section, we shall consider one dimensional motion, that is, motion along a straight line.



8.2.1 Terminology

Scalars are quantities which are fully described by magnitude alone.

Vectors are quantities decribed by magnitude and direction.

Displacement x is a vector quantity which refers to "how far out of place an object is"; it is the object's change in position.

Speed is a scalar quantity which refers to "how fast" an object is moving.

- A fast moving object has a high speed while a slow moving object has a low speed.
- An object with no movement at all has a zero speed.
- average speed $= \dfrac{\text{distance travelled}, x}{\text{time of travel}, t}$

- Instantaneous speed $= \frac{dx}{dt}$

Velocity is a vector quantity which refers to "the rate at which an object changes its position."

- when evaluating the velocity of an object, one must keep track of direction.
- average velocity $= \dfrac{\text{displacement}, x}{\text{time}, t}$
- instantaneous velocity, $\frac{dx}{dt}$ – the velocity of a particle at any instant of time.

Acceleration is a vector quantity which is defined as "the rate at which an object changes its velocity"

- An object is accelerating if it is changing its velocity.
- acceleration $= \frac{dv}{dt} = \frac{dX}{dt^2}$

Example 35. The position of an object after moving horizontally is given by $x = t^3 - 6t^2 - 15t + 40$ with x meter as coordinate with respect to the origin at time t second. Find

(a) time the object stop moving,

(b) position and distance the object moved within the above time,

(c) acceleration of the object within the above time,

(d) distance during time $4 \leq t \leq 6$s.

Solution.

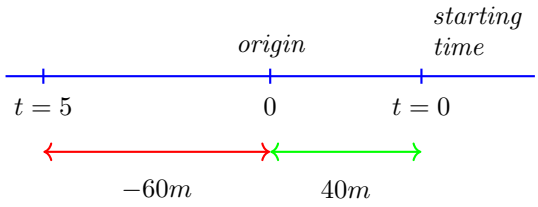
(a) When the object stops, velocity is 0, so $\frac{dx}{dt} = 0$. *If* $x = t^3 - 6t^2 - 15t + 40$ *then*

$$\begin{aligned} \frac{dx}{dt} &= 3t^2 - 12t - 15 = 0 \\ t^2 - 4t - 5 &= 0 \\ (t-5)(t+1) &= 0 \\ t &= 5,\ t = -1. \end{aligned}$$

Therefore, the object stops moving after 5s.

(b) *Position and distance the object moves within the above time At* $t = 5$, *distance travelled is* $x = 5^3 - 6(5)^2 - 15(5) + 40 = -60\text{m}$.

The object position at time $t = 0$ *is* $x = 0 - 0 - 0 + 40 = 40\text{m}$.



So the distance traveled by object in 5s *is* $60\text{m} + 40\text{m} = 100\text{m}$.

(c) *acceleration of the object within the above time*

$$acceleration = \frac{dv}{dt} = \frac{d^2}{dt^2} = 6t - 12,$$

at $t = 5$,

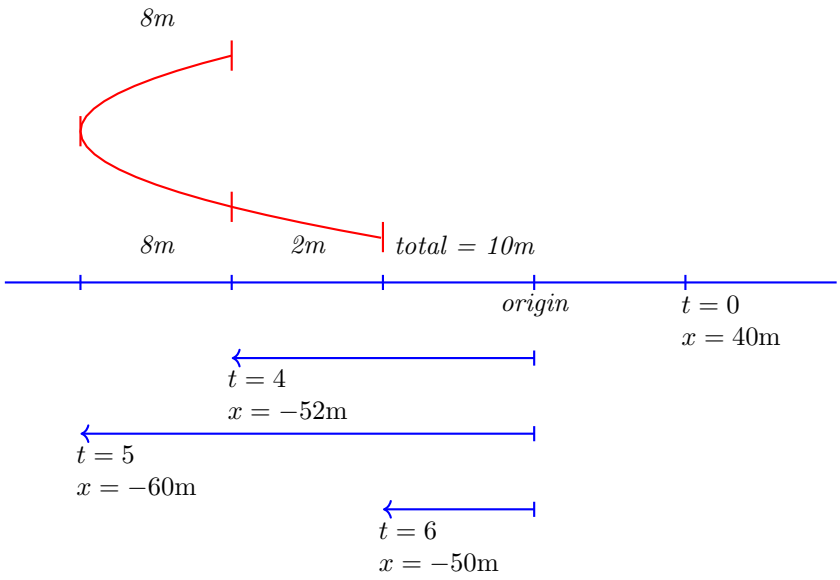
$$v = \frac{d^2x}{dt^2} = a = 6(5) - 12 = 18\text{ms}^{-2}.$$

(d) *distance during time* $4 \leq t \leq 6\text{s}$.

$$\begin{aligned} v = \frac{dx}{dt} &= 3t^2 - 12t - 15 \\ &= 3(t+1)(t-5). \end{aligned}$$

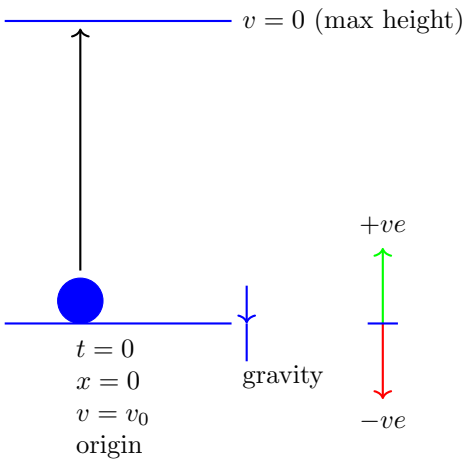
Object positions

$$\begin{aligned} &t = 4,\ x(4) = -52\text{m}, \\ &t = 5,\ x(5) = 60\text{m}, \\ &t = 6,\ x(6) = -50\text{m}. \end{aligned}$$



Therefore, the total distance is $8 + (8 + 2) = 18$m.

8.3 Vertical motion under gravity



Example 36. An object is thrown straight upwards with an initial velocity 15ms^{-1}. Acceleration under gravity is 9.8ms^{-2}. Calculate

(a) time taken to reach maximum height,

(b) the maximum height from the original position,

(c) time taken to return to its original position,

(d) velocity when it return to its original position.

Solution.

(a) Acceleration is $\frac{dv}{dt} = -g$*, a separable equation. It follows that* $v = -gt + c$.

Initial velocity is $v_0 = 15\text{ms}^{-1}$*, that is* $v(0) = 15$*. This implies* $c = 15$*, so*

$$v = -gt + 15. \tag{8.1}$$

When maximum height is attained, $v = 0$*,*

$$\begin{aligned} 0 &= -9.8t + 15 \\ t &= \frac{15}{9.8} = 1.53\text{s}. \end{aligned}$$

(b) The equation (8.1) can be written as

$$\begin{aligned} \frac{dx}{dt} &= -gt + 15 \\ \int dx &= \int -gt + 15\ dt \\ x &= -\frac{gt^2}{2} + 15t + c. \end{aligned}$$

At $t = 0$*,* $x = 0$*, this implies* $c = 0$ *so*

$$x = -\frac{gt^2}{2} + 15t. \tag{8.2}$$

The time taken to reach maximum height is $t = 1.53\text{s}$*, so the maximum height is* $x = -\frac{9.8(1.53)^2}{2} + 15(1.53) = 11.48\text{m}$.

Alternatively,

$$\begin{aligned} \frac{dv}{dt} &= \frac{dv}{dx} \cdot \frac{dx}{dt} = v\frac{dv}{dx} \\ \textit{since} \quad \frac{dv}{dt} &= -g, \\ \textit{then} \quad v\frac{dv}{dx} &= -g, \quad \textit{a separable equation} \\ \int v\ dv &= \int -g\ dx \\ \frac{v^2}{2} &= -gx + c. \end{aligned}$$

When $x = 0$, $v = 15$ *so* $\frac{15}{2} = -9.8(0) + c$ *giving* $c = 112.5\text{m}$. *Therefore,*

$$\frac{v^2}{2} = -gx + 112.5.$$

At the maximum height, $v = 0$, *we have* $0 = -9.8x + 112.5$, *which implies* $x = 11.48\text{m}$.

(c) From (8.2),

$$x = \frac{gt^2}{2} + 15t = t\left(\frac{gt}{2} + 15\right).$$

When $x = 0$, $t = 0\text{s}$ *and* $t = 3.06\text{s}$. *Therefore, the time taken to return to its original position is* $t = 3.06\text{s}$.

(d) From (8.1), *when* $t = 3.06\text{m}$,

$$v = -9.8(3.06) + 15 = -14.99 \approx -15\text{ms}^{-1}.$$

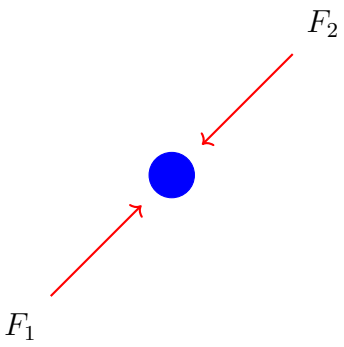
Object moves with velocity 15ms^{-1} *downwards.*

8.4 Newtonian mechanics

- Mechanics is the study of the motion of objects and the effect of forces acting on those objects.
- Deals with the motion of ordinary objects, that is, object that are large compared to an atom and slow moving compared with the speed of light.
- A model for Newtonian mechanics can be based on **Newton's laws of motion**:

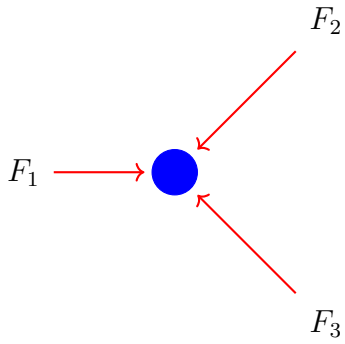
8.4.1 Newton's laws of motion

1. Let F_1, F_2 be externa1 forces.



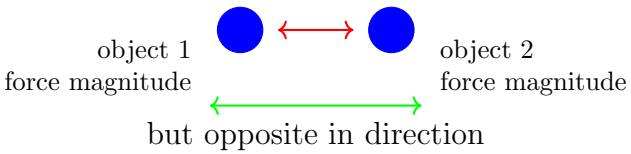
If $F_1 + F_2 = 0$, then it moves with a constant velocity.

2. when a body is subjected to one or more externa1 forces, say F_1, F_2, F_3,



then $\frac{dp}{dt} = F_1 + F_2 + F_3$.

3. when one body interacts with a second body,



8.4.2 Procedure for Newtonian models

(a) Determine all relevant forces acting on the object being studied. (It is helpful to draw a simple diagram of the object that depicts these forces)

(b) Choose an appropriate axis or coordinate system in which to represent the motion of the object and the forces acting on it. Keep in mind that this coordinate system must be an inertial reference frame.

(c) Apply Newton's second law

$$\frac{dp}{dt} = F(t, x, v), \quad p(t) = mv(t)$$

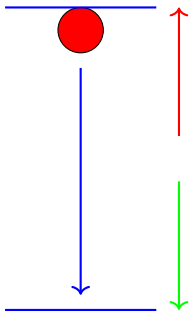
or

$$\frac{dp}{dt} = \frac{d(mv)}{dt} = m\frac{dv}{dt} = ma = F(t, x, v),$$

whichever is appropriate, to determine the equations of motion for the object.

Example 37. An object of mass 1 kg is released and falls under the influence of gravity. The force due to air resistance is $\frac{7}{x+1}$ with x being its distance from the origin. Find the velocity of the object when it falls to $x = 6.4$m. [Use $g = 10\text{ms}^{-2}$]

Solution. *At the beginning,* $t = 0, x = 0, v = 0$. *Mass,* $m = 1$kg.



$F_1 = -\frac{7}{x+1}$

$F_1 + F_2 = mg - \frac{7}{x+1}$

$F_2 = mg$

From Newton's Second Law of Motion, $F = ma$,

$$\begin{aligned}
m\frac{dv}{dt} &= mg - \frac{7}{x+1} \\
\frac{dv}{dt} &= g - \frac{7}{x+1}, \quad \textit{if } m = 1 \\
\frac{dv}{dx} \cdot \frac{dx}{dt} &= g - \frac{7}{x+1} \\
\frac{dv}{dx} \cdot v &= g - \frac{7}{x+1} \\
\int v\, dv &= \int \left(g - \frac{7}{x+1}\right)\, dx \\
\frac{v^2}{2} &= gx - 7\ln|x+1| + c.
\end{aligned}$$

When $x = 0$, $v = 0$ *so*

$$\begin{aligned}
0 &= 0 - 7\ln|1| + c \\
c &= 0,
\end{aligned}$$

which implies

$$\frac{v^2}{2} = gx - 7\ln|x+1|.$$

If $x = 6.4$m, *then*

$$\begin{aligned}
\frac{v^2}{2} &= 10(6.4) - 7\ln|7.4| \\
v &= 9.99 \approx 10\text{ms}^{-1}.
\end{aligned}$$

Example 38. An object of mass m kg is given a downward velocity v_0 and is then allowed to fall under the insfluence of gravity. Assume that the influence of gravity is constant and the force due to air resistance is proportional to the velocity of the object. Determine the equation of the motion of the object.

Solution. *From Newton's Second Law of Motion, $F = ma$,*

$$\begin{aligned} m\frac{dv}{dt} &= mg - kv \\ \int \frac{m}{mg-kv}dv &= \int dt \\ -\frac{m}{k}\ln|mg-kv| &= t + c \\ \ln|mg-kv| &= -\frac{kt}{m} + \frac{ck}{m} \\ |mg-kv| &= e^{-\frac{kt}{m}+\frac{ck}{m}} \\ &= Ae^{-\frac{kt}{m}}, \quad \textit{where } A = e^{\frac{ck}{m}}. \end{aligned}$$

At $t = 0$, $v(0) = v_0$, so $mg - kv_0 = A$, and

$$\begin{aligned} mg - kv &= e^{-kt/m}(mg - kv_0) \\ -kv &= e^{-kt/m}(mg - kv_0) - mg \\ v &= \frac{mg}{k} - \frac{e^{-kt/m}}{k}(mg - kv_0) \\ v(t) &= mg + e^{-kt/m}\left(-v_0 - \frac{mg}{k}\right). \end{aligned}$$

Now

$$\begin{aligned} v(t) = \frac{dx}{dt} &= mg + e^{-kt/m}\left(-v_0 - \frac{mg}{k}\right) \\ \int dx &= \int mg + e^{-kt/m}\left(-v_0 - \frac{mg}{k}\right)\, dt \\ x &= \left(\frac{mg}{k}\right)t - \frac{me^{-kt/m}}{k}\left(v_0 - \frac{mg}{k}\right) + c. \end{aligned}$$

When $t = 0$, $x = 0$, so

$$\begin{aligned} 0 &= 0 - \frac{me^0}{k}\left(v_0 - \frac{mg}{k}\right) + c \\ c &= \frac{m}{k}\left(v_0 - \frac{mg}{k}\right). \end{aligned}$$

Then

$$\begin{aligned} x(t) &= \left(\frac{mg}{k}\right) t - \frac{me^{-kt/m}}{k}\left(v_0 - \frac{mg}{k}\right) + \frac{m}{k}\left(v_0 - \frac{mg}{k}\right) \\ &= \left(\frac{mg}{k}\right) t + \left(\frac{m}{k}\right)\left(1 - \frac{me^{-kt/m}}{k}\right)\left(v_0 - \frac{mg}{k}\right). \end{aligned}$$

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